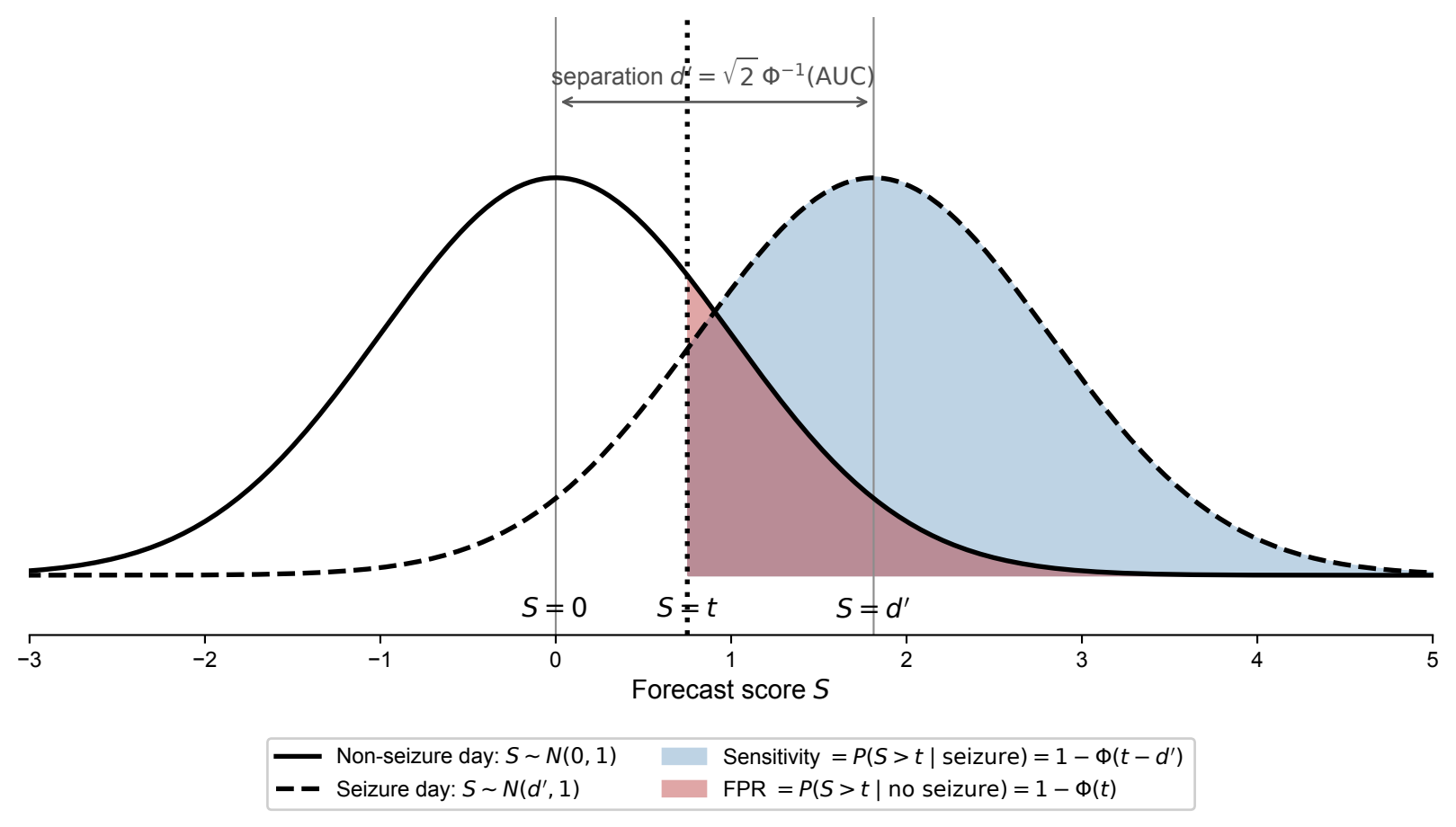
separation $d' = \sqrt{2}\,\Phi^{-1}(\mathrm{AUC})$
$S = 0$
$S = t$
$S = d'$
−3
−2
−1
0
1
2
3
4
5
Forecast score S
Non-seizure day: $S \sim N(0, 1)$
Seizure day: $S \sim N(d', 1)$
Sensitivity $= P(S > t \mid \text{seizure}) = 1 - \Phi(t - d')$
FPR $= P(S > t \mid \text{no seizure}) = 1 - \Phi(t)$